Mock DS — Python for Data Science

/20

Part 1: **theory, on paper** · Chapters 0, 1, 2 · Duration: 1 h · No documents, no computer

Name: _____ Group: _____ Date: _____

Part I — Multiple choice (6 pts · 0.5 per question · one correct answer)

Q1. Which phase of the software lifecycle checks that the program works correctly?

☐ a) Design ☐ b) Testing
☐ c) Deployment ☐ d) Maintenance

Q2. Which one is NOT one of the 5 components of an Information System?

☐ a) People ☐ b) Process
☐ c) Internet ☐ d) Data

Q3. The user types `20` in `age = input("Age: ")`. What is `type(age)`?

☐ a) int ☐ b) str
☐ c) float ☐ d) it depends on what is typed

Q4. `17 // 5` and `17 % 5` give:

☐ a) 3.4 and 2 ☐ b) 3 and 2
☐ c) 3 and 3.4 ☐ d) 2 and 3

Q5. Which object cannot be modified after its creation?

☐ a) list ☐ b) dict
☐ c) tuple ☐ d) NumPy array

Q6. `np.array([2, 4.0, True]).dtype` is:

☐ a) int64 ☐ b) bool
☐ c) float64 ☐ d) <U32

Q7. `np.arange(12).reshape((3, 5))`:

☐ a) returns a 3×5 array ☐ b) returns a 5×3 array
☐ c) raises a ValueError ☐ d) pads the array with zeros

Q8. Which pandas command shows the dtypes AND the number of non-null values per column?

☐ a) df.describe() ☐ b) df.info()
☐ c) df.head() ☐ d) df.shape

Q9. A DataFrame has 12,060 rows and 10 columns. `df.shape` is:

☐ a) 120600 ☐ b) (10, 12060)
☐ c) (12060, 10) ☐ d) [12060, 10]

Q10. Which line correctly keeps the Tunis orders with more than 3 units?

☐ a) df[df.city == "Tunis" and df.q > 3] ☐ b) df[(df["city"] == "Tunis") & (df["q"] > 3)]
☐ c) df[df["city"] == "Tunis" & df["q"] > 3] ☐ d) df.loc["city" == "Tunis"]

Q11. `pd.to_datetime(s, errors="coerce")` on the value `"2025-13-40"`:

☐ a) raises an error ☐ b) returns NaT
☐ c) returns 2025-12-31 ☐ d) keeps the string

Q12. With the default index (0, 1, 2…), how many rows does `df.iloc[0:3]` return? And `df.loc[0:3]`?

☐ a) 3 and 3 ☐ b) 3 and 4

☐ c) 4 and 3 ☐ d) 2 and 3

Q	1	2	3	4	5	6	7	8	9	10	11	12
Answer												

Part II — What does this code print? (6 pts · 1 per snippet)

Write the exact output. If the code raises an error, give the name of the error.

P1.

```
x = "4"
y = 2
print(x * y, int(x) * y, int(x) / y)
```

P2.

```
total = 0
for i in range(2, 10, 3):
    total += i
    print(i, end=" ")
print("|", total)
```

P3.

```
lst = [10, 20, 30, 40, 50]
lst.append(60)
print(lst[1:4], lst[-2], len(lst))
print(lst[::2])
```

P4.

```
d = {"Tunis": 3, "Sfax": 1}
d["Sousse"] = 2
d["Tunis"] += 1
for k, v in d.items():
    if v > 1:
        print(k, v)
```

P5.

```
import numpy as np
a = np.array([[1, 2, 3], [4, 5, 6]])
print(a.shape, a.size, a.sum(axis=0), a[1, 2])
print(a[a > 2])
```

P6.

```
import pandas as pd
df = pd.DataFrame({"city": ["Tunis", " tunis", "Sfax", "Sfax"],
                   "q": [2, None, 5, 5]})
print(df.shape, df["q"].isna().sum(), df.duplicated().sum())
df["city"] = df["city"].str.strip().str.title()
print(df["city"].nunique(), df["q"].dtype, df["q"].mean())
```

Part III — Short questions (4 pts · 1 per question)

S1. The column `quantity` of a CSV contains whole numbers, but pandas loaded it as `float64` (1.0, 2.0…). Why? What must you do before `astype("int64")`?

S2. Give two different treatments for missing values with an example of when each is appropriate. Why must we not replace all missing values by 0?

S3. Explain the difference between `df.loc` and `df.iloc`. Give an example where they return different rows.

S4. What does `orders.groupby("product_id")["unit_price"].transform("median")` return, compared with `.median()`? Why is it useful for cleaning?

Part IV — Write the code by hand (4 pts)

C1. Write a function `grade(score)` that returns `"Excellent"` if score ≥ 16, `"Good"` if ≥ 12, `"Pass"` if ≥ 10 and `"Fail"` otherwise. Then, for `scores = [8.5, 12, 17.25, 10, 9.75, 14]`, print each score with its grade and print how many students passed (score ≥ 10) and the class average. 2 pts

C2. A DataFrame `orders` has the columns `order_id, product_id, city, quantity, unit_price`. Write the code to: (a) remove the duplicated rows; (b) clean `city` (spaces and upper/lower case); (c) keep only the lines with at most 10 units; (d) fill the missing `unit_price` with the median price of the same product; (e) create `revenue` = quantity × unit_price; (f) display the total revenue per city, from highest to lowest; (g) export the result to `orders_clean.csv` without the index. 2 pts